

ICT600 CORRECTED STUDENT SCHEME

February 2025

Question-ordered model answers and marking guide

Scope	February 2025 final examination
Arrangement	Answers follow the original paper from Question 1 through Part B. Coding, diagrams, and written responses are integrated once at their correct positions.
Assessment	100 marks: Part A 70, Part B 30
Status	Corrected student reference; equivalent technically correct answers may receive credit.

How to use this scheme

Question-order rule

Attempt the paper first. Then check the matching answer in this document. Each answer appears once, in the same order as the question paper.

- Equivalent correct wording, diagrams, algorithms, and code may receive full credit when they meet the stated requirements.
- For coding, award method marks when a minor syntax error does not invalidate an otherwise correct solution.
- For design or illustration questions, the labelled diagram is the model answer; prose alone is not treated as a complete design.

Part A

Question 1(a)

8 marks

Describe Intranet, Extranet, Internet, and LAN.

Mark allocation

Awardable element	Marks
Intranet	2
Extranet	2
Internet	2
LAN	2
Total	8

Complete model answer

- Intranet: a private organisational network for authorised internal users.
- Extranet: controlled access to selected organisational resources for authorised external parties.
- Internet: the global network of interconnected networks using standard protocols.
- LAN: a network covering a limited area such as a room, building, or campus section.

Question 1(b)

10 marks

Identify four cloud applications used for teaching during MCO/PKP, with their uses.

Mark allocation

Awardable element	Marks
First application and use	2.5
Second	2.5
Third	2.5
Fourth	2.5
Total	10

Complete model answer

- Any four relevant applications such as UFuture, Google Classroom, Microsoft Teams, Google Meet, Google Drive, or OneDrive, each paired with an accurate teaching use.

Question 2(a) - student table

6 marks

Reproduce Table 1 using valid HTML table elements.

Mark allocation

Awardable element	Marks
Table structure	1
Header row	1
Three complete data rows	3
Valid nesting	1
Total	6

Complete model answer

```
<table border="1">
  <tr><th>Namape lajar</th><th>Namabapa</th><th>Umur</th></tr>
  <tr><td>Ahmad</td><td>Razman</td><td>51</td></tr>
  <tr><td>Mohamad</td><td>Nizar</td><td>55</td></tr>
  <tr><td>Abdul</td><td>Syazali</td><td>47</td></tr>
</table>
```

Question 2(b) - styled HTML page

6 marks

Organise the required powderblue centred page.

Mark allocation

Awardable element	Marks
Valid HTML and text	2
Body styling	2
Blue h1	1
Red paragraph	1
Total	6

Complete model answer

```
<!doctype html>
<html lang="en">
<head>
  <meta charset="utf-8">
  <title>Web Application and Technology</title>
  <style>
    body { background-color: powderblue; text-align: center; }
    h1 { color: blue; }
    p { color: red; }
  </style>
</head>
<body>
  <h1>Web Application and Technology</h1>
  <p>Explore web technology and application</p>
</body>
</html>
```

Question 3(a) - JavaScript variables

8 marks

Assign the printed name, class, year and CGPA, then display them.

Mark allocation

Awardable element	Marks
Four correct variables/values	4
Four matching outputs	4
Total	8

Complete model answer

```
<!doctype html>
<html lang="en"><body>
<p id="name"></p><p id="class"></p><p id="year"></p><p id="cgpa"></p>
<script>
const studentName = "Shamal bin Aidi";
const className = "Al-Khawarizmi";
const year = 8;
const cgpa = 3.55;
document.getElementById("name").textContent = "Name: " + studentName;
document.getElementById("class").textContent = "Class: " + className;
document.getElementById("year").textContent = "Year: " + year;
document.getElementById("cgpa").textContent = "CGPA: " + cgpa;
</script>
</body></html>
```

Question 3(b) - displayUserName

10 marks

Call displayUserName('Sharaf') and display Name: Sharaf.

Mark allocation

Awardable element	Marks
Function	2
Parameter	2
Name prefix	2
DOM output	2
Correct call	2
Total	10

Complete model answer

```
<p id="result"></p>
<script>
function displayUserName(userName) {
  document.getElementById("result").textContent = "Name: " + userName;
}
displayUserName("Sharaf");
</script>
```

Expected output / behaviour

Name: Sharaf

Marker note

The historical official answer copied Aidan from another paper. Sharaf is required here.

Question 4(a) - family table

6 marks

Create the complete Table 2 with the spanning title row.

Mark allocation

Awardable element	Marks
Table structure	1
Spanning title	2
Village headings	1
Complete values	2
Total	6

Complete model answer

```
<table border="1">
  <tr><th colspan="2">Bilangan Kanak-kanak dalam Keluarga Bawah 7 tahun 2023</th></tr>
  <tr><th>Kampung Tin</th><th>Kampung Changkat</th></tr>
  <tr><td>300 orang</td><td>200 orang</td></tr>
  <tr><td>150 keluarga</td><td>100 keluarga</td></tr>
</table>
```

Question 4(b) - HTML and JavaScript page

6 marks

Produce the illustrated page and a working click interaction.

Mark allocation

Awardable element	Marks
Heading, message and button	3
Event	1
DOM update	2
Total	6

Complete model answer

```
<!doctype html>
<html lang="en"><body>
  <h1>What Can You Do?</h1>
  <p id="message">Hello You!</p>
  <button type="button" id="change">click me!</button>
  <script>
    document.getElementById("change").addEventListener("click", () => {
      document.getElementById("message").textContent = "You can use JavaScript!";
    });
  </script>
</body></html>
```

Marker note

The historical official wording was copied from a different paper; mark against this paper's Figure 2.

Question 5(a) - PHP city array

4 marks

Use a multidimensional array to display the two city rows.

Mark allocation

Awardable element	Marks
Nested array	2
Complete output	2
Total	4

Complete model answer

```
<?php
$cities = [
    ['LN11', 'London'],
    ['OX47', 'Oxford'],
];
?>
<table border="1">
<?php foreach ($cities as $city): ?>
    <tr><td><?= $city[0] ?></td><td><?= $city[1] ?></td></tr>
<?php endforeach; ?>
</table>
```

Question 5(b) - exact PHP trace

6 marks

Produce the exact output of the printed PHP string operations.

Mark allocation

Awardable element	Marks
Original	1
Uppercase	1
Lowercase	1
Replacement	1
Substring	1
strpos result	1
Total	6

Complete model answer

```
How can I cook rice in a clean round pot
HOW CAN I COOK RICE IN A CLEAN ROUND POT
how can i cook rice in a clean round pot
How round I cook rice in a clean round pot
w roun
bool(false)
```

Part B

Part B Question 1 - all-required PHP registration

15 marks

Provide complete form processing where full name, email, password, birthdate and gender are all required.

Mark allocation

Awardable element	Marks
Complete form	4
POST handling	2
All five fields validated	4
Success/error messages	2
Safe output	2
Valid structure	1
Total	15

Complete model answer

```
<?php
$values = ['fullname' => '', 'email' => '', 'birthdate' => '', 'gender' => ''];
$message = '';

function e(string $value): string {
    return htmlspecialchars($value, ENT_QUOTES, 'UTF-8');
}

if ($_SERVER['REQUEST_METHOD'] === 'POST') {
    foreach ($values as $key => $unused) {
        $values[$key] = trim($_POST[$key] ?? '');
    }
    $password = $_POST['password'] ?? '';
    if ($values['fullname'] === '' || $values['email'] === '' || $password === '' ||
        $values['birthdate'] === '' || $values['gender'] === '') {
        $message = 'Unsuccessful: enter all required information.';
    } else {
        $message = 'Successful signup for ' . $values['fullname'] . '.';
    }
}
?>
<!doctype html>
<html lang="en"><head><meta charset="utf-8"><title>User Signup</title></head>
<body>
<h1>User Signup</h1>
<form method="post">
    <label>Full name* <input name="fullname" value="<?= e($values['fullname']) ?>" required></label><br>
    <label>Email* <input type="email" name="email" value="<?= e($values['email']) ?>" required></label><br>
    <label>Password* <input type="password" name="password" required></label><br>
    <label>Birthdate <input type="date" name="birthdate" value="<?= e($values['birthdate']) ?>"></label><br>
    <label><input type="radio" name="gender" value="M"> Male</label>
    <label><input type="radio" name="gender" value="F"> Female</label><br>
    <button type="submit">Signup</button>
</form>
<?php if ($message !== ''): ?><p><?= e($message) ?></p><?php endif; ?>
</body></html>
```

Part B Question 2

15 marks

Recommend a checkout-payment-page design using a wireframe.

Mark allocation

Awardable element	Marks
Header and identity	2
Navigation and page name	2
Billing information	2
Delivery information	2
Payment information	2
Order summary	2
Purchase confirmation	2
Footer and coherent layout	1
Total	15

Complete model answer

- The model below shows a complete acceptable layout. Equivalent arrangements receive credit when all required functional areas are present.

HEADER: LOGO | SEARCH | ACCOUNT / CART

NAVIGATION > CART > CHECKOUT > PAYMENT

BILLING ADDRESS

DELIVERY ADDRESS

DELIVERY METHOD

ORDER SUMMARY
Items | Delivery | Total

PAYMENT METHOD
Card / Online Banking

PLACE ORDER

FOOTER: CONTACT | DELIVERY | RETURNS | PRIVACY